

Reza Mahjourian

Website, Google Scholar

RESEARCH PROFILE

- ❖ Research scientist specializing in foundation models for robotics and autonomous systems, with a track record of translating frontier research into deployed AI systems.
- ❖ Expertise in large-scale world modeling, multi-agent prediction, and applying large-scale vision and vision-language models to robotics, autonomous driving, and safety-critical systems.
- ❖ Author of 13 patents and publications in CVPR, ECCV, ICRA, IROS, and AAAI with 2500+ citations.
- ❖ Selected contributions include saliency-guided foundation models, occupancy flow fields for motion prediction, scalable scenario generation for autonomous systems, sample-efficient reinforcement learning for robot control, and self-supervised learning of depth and ego-motion.

SKILLS

A.I. RESEARCH

- Foundation Models
- Robotics
- Computer Vision
- Multi-Agent Prediction
- Reinforcement Learning
- Evolutionary Strategies

EDUCATION

UNIV. OF TEXAS AT AUSTIN

PHD IN COMPUTER SCIENCE

GPA: 4.0

SHARIF UNIV. OF TECH.

BS IN COMPUTER ENG.

Tehran, Iran

OPEN SOURCE

- ❖ **GameGraph** [↗](#): Studies impact of domain stochasticity and ergodicity on reinforcement learning.
- ❖ **Discovery** [↗](#): Evolutionary feature discovery for reinforcement learning.
- ❖ **OpenNERO** [↗](#): Platform for A.I. research and education.

EXPERIENCE

WAYMO RESEARCH

Jul 2023 - Present | Staff Research Scientist

Jul 2019 - Jul 2023 | Senior Research Scientist

- Led development of a saliency-guided vision preprocessing framework enabling efficient fine-tuning of Gemini multimodal foundation models for safety-critical autonomous driving tasks such as detection and segmentation, improving long-tail object detection by 5–34% and segmentation accuracy by 10–16%.
- Led research on scalable occupancy and trajectory prediction. Established occupancy modeling as a shared representation across prediction, simulation, and planner evaluation pipelines, and led cross-functional efforts with Perception and Simulation teams to deliver two production systems: the onboard Occlusion Prediction System and a Scenario Generation pipeline for synthesizing novel driving scenarios.
- Founded and organized the Waymo Open Dataset Occupancy & Flow Prediction and Scenario Generation challenges at CVPR (**2022, 2024, 2025**), engaging 20+ academic and industry teams and catalyzing 15+ follow-up peer-reviewed publications across academia and industry.

GOOGLE BRAIN ROBOTICS | STUDENT RESEARCHER

Sep 2017 - Nov 2018 | Mountain View, CA

- Developed sample-efficient reinforcement learning methods for robot table tennis using self-play (**ArXiv 2018, Website** [↗](#)). The hierarchical and modular policy design was later adopted in a multi-year project achieving human-level competitive robot table tennis (**RSS 2023, ICRA 2025, Videos** [↗](#)).
- Unsupervised learning of object depth and motion from raw monocular videos. **AAAI 2019, CVPR 2019**, patent, **Website** [↗](#), **Google AI Blog** [↗](#).
- Future semantic segmentation using 3D structure, **ECCV 2018**, patent.
- Added computer vision library functions to **TensorFlow** [↗](#).

GOOGLE BRAIN ROBOTICS | RESEARCH INTERN

May 2017 - Aug 2017 | Mountain View, CA

- Developed a novel self-supervised learning method for depth and ego-motion from monocular videos. **CVPR 2018, Website** [↗](#), patent, patent, patent.
- Developed custom TensorFlow op for aligning 3D point clouds during training.

GOOGLE BRAIN | RESEARCH INTERN

May 2016 - Aug 2016 | Mountain View, CA

- Geometry-based next frame prediction from monocular video.
- IEEE Intelligent Vehicles Symposium 2017 **Paper** [↗](#), patent, patent.

GOOGLE | SOFTWARE ENGINEERING INTERN

Jun 2015 - Aug 2015 | Mountain View, CA

- Designed large-scale neural models for predicting in-app purchases using full Google Play historical data. Achieved measurable lift over production baselines.

TEAMUP [↗](#) | START-UP LEAD DEVELOPER

Jul 2009 - Dec 2009 | London, UK

- Created a Django web app for managing fitness and sports businesses.
- Led the project from inception to beta release **Website** [↗](#)

TELEMEDIA | SOFTWARE ENGINEER

Sep 2004 - Dec 2005 | Paris, France

- Created a web app for a French network operator (KDDI) enabling customers to query their call data records.
- Created a PostgreSQL database pooling call detail records from telecom devices. Optimized the database and API for very fast queries.
- Developed a Python backend to process and fulfill mobile content orders placed via web services and via SMPP over the GPRS network.

VEESTA WORLD CO | SOFTWARE ENGINEER

Sep 2003 - Sep 2004 | Tehran, Iran

- Implemented a SCADA communication protocol for an industrial RTU, running on the PSOS real-time OS. RTUs were deployed in a system monitoring the national power network.

SCIENCE AND ARTS FOUNDATION | SOFTWARE ENGINEER

Aug 1999 - Aug 2003 | Tehran, Iran

- Led the technical operations of a non-profit organization based in Sharif University of Technology, donating computers and internet access to schools in underprivileged areas.
- Managed Linux servers for a network of about a 100 schools connected to Sharif University.
- Created a web portal offering educational content and services to teachers and students.
- Facilitated participation of teachers and students in international collaborative projects. Talks:
 - “The Experiences of iEARN in Iran”, Osaka, Japan, 2003
 - “Making a School Intranet Using Free and Open-Source Software”, Moscow, Russia, 2002
 - “The Activities of Science and Arts Foundation in Iran”, Cape Town, South Africa, 2001

KANOON INFORMATION SYSTEMS | START-UP CO-FOUNDER

May 1997 - Sep 2002 | Tehran, Iran

- Created and maintained a web app for TakTaz Motors, enabling the sales office, remote factory, and distributed dealers to collaboratively manage inventory, wholesale orders, shipments, and registration documents.
- Led development of a database application for a major automaker (Kaveh Khodro) managing and tracking orders placed with parts suppliers, featuring comprehensive progress reports.
- Developed a multiple choice test evaluation package for educational institutions, offering reports on test taker performance and progress over time.

IDEH-NEGAR SOFTWARE | SOFTWARE DEVELOPER

Dec 1995 - Apr 1997 | Tehran, Iran

- Programmed custom database applications for company clients.
- Created a customizable automated answering system serving customers over the phone.
- Programmed a library for visualizing metrics for a training a neural network doing OCR over Persian language text.